



Higher autism trait severity in children associates with diminished integration of angry prosody in semantic processing

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ARTICLE INFO

Keywords:

Autism traits
Emotional prosody
Semantic processing
Speech
Communication
Event-related potentials

ABSTRACT

Background: Previous studies examined group differences of individuals with and without autism. This study investigated individual differences in automatic integration of emotional prosody during semantic processing at the behavioral and neural levels among children using a continuous measure of autism trait severity.

Method: Event-related potentials (ERPs) of 46 children, 21 of them diagnosed with ASD, were measured. Children determined the semantic valence of spoken words, and ignored the emotional prosody, which was either congruent or incongruent with the semantic valence.

Results: Independent of congruency, higher autism trait severity and Intelligence Quotient (IQ) were associated with faster reaction times (RTs), while higher IQ and language ability were associated with higher accuracy. The difference between congruent and incongruent conditions (congruency effect) indicates automatic integration of emotional prosody in semantic processing. In ERPs, higher autism trait severity and higher IQ were associated with smaller congruency effects on P2 amplitudes for angry (but not happy) prosodic stimuli.

Conclusion: The findings suggest a stronger tendency to ignore angry prosodies among children with higher autism trait severity. This has implications for educational practices and parent counseling regarding communication with children with high autism trait severity.

1. Introduction

Emotional expressions are shared through verbal and nonverbal channels, but accurate interpretation of others' emotions often relies on integration across emotion-carrying channels. Previous studies have revealed deficits in the ability to process verbally expressed emotions among individuals with autism spectrum disorder (ASD) (Lartseva et al., 2015; Leipold et al., 2023), a disorder

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characterized by social communication deficits and repetitive behavioral patterns (American Psychiatric Association, 2013). Other ASD studies have examined nonverbal signals such as emotional prosody, expressed via pitch, intensity, and duration of speech utterances (Scherer et al., 1991), with inconclusive results. While some found deficits in recognition of emotional prosody in adults (Golan et al., 2007; Rutherford et al., 2002), adolescents and children with ASD (Leipold et al., 2023; Lindner & Rosén, 2006), others found intact emotional prosody recognition abilities (Boucher et al., 2000; Grossman et al., 2010; Lyons et al., 2014) or subtle deficits (Brennand et al., 2011; Chevallier et al., 2011) in children and adolescents with ASD.

Most studies on ASD examine emotions in a single, verbal or nonverbal channel. Few have investigated multimodal emotional information. Charbonneau and colleagues found smaller bimodal audio-visual advantages (compared to unimodal conditions) for processing emotions in ASD individuals, implying an integration deficit (Charbonneau et al., 2013). In contrast, Brewer and colleagues found similar integration of emotional facial and body cues in adults with and without ASD (Brewer et al., 2017). Here, we focused on the automatic integration of emotional prosody and semantic word meaning in children with different autism trait severity, at behavioral and neural levels. Previously, Singh and Harrow (2014) examined the processing of emotional prosody and semantics in typically developing (TD) and children with ASD who repeated spoken words and made judgments on emotional prosody and speaker sex. While the ASD and TD groups performed similarly in judging prosody and sex, only the TD group showed delayed responses during the word repetition task if prosody and meaning were incongruent, indicating impaired emotional prosodic-semantic integration in ASD.

In our study, the stimuli were spoken words with either congruent or incongruent emotional prosodies. An example of a congruent stimulus is the word “love” spoken with a happy voice, while for incongruent stimulus, the word “praise” spoken with an angry voice. Participants were required to judge the meaning of words (whether it is positive or negative) while ignoring the task-irrelevant emotional prosodies. The performance difference between incongruent and congruent conditions (the congruency effect) reflects the automatic integration of emotional prosody. In addition to behavioral measures, the high temporal resolution of ERP revealed the precise time frames of cognitive activities involved in processing semantic-prosodic incongruency, and may provide important cues about individual differences in cognitive mechanism underlying emotional speech processing. We also included several control variables, which may confound integration effects. There is common comorbidity of ASD (Loucas et al., 2008) with developmental language disorders and intellectual disability, which contributes to deficits in the processing of emotional speech. In addition, sex differences in terms of a female advantage in the integration of prosodic and semantic cues are well documented (Schirmer et al., 2002). Hence, we addressed the role of autism trait severity in integrating emotional prosody in speech while controlling for language ability, intelligence, and sex.

The functional locus of emotional prosodic-semantic integration within the neurocognitive system can be assessed with ERPs. In a group of neurotypical adults varying in the degree of alexithymia, Wang et al. (2021) observed early ERP effects of emotional speech on P2 amplitudes (Wang et al., 2021), which are negatively associated with the level of attention in processing emotions (Kanske et al., 2011). Alexithymia is a trait characterized by impairments to identify and describe emotions. High alexithymia group showed larger P2 amplitudes and poorer performance in judging sad and angry prosodies, and ignored simultaneously presented faces with congruent or incongruent emotional expressions. As there is high comorbidity of ASD and alexithymia, we investigated whether ASD severity also modulates the congruency effect of semantics and prosody on P2. Another ERP component, N2, is generally found to be related to conflict monitoring (van Veen & van Veen & Carter, 2002) and response inhibition (Nieuwenhuis et al., 2003) in both adults and children (Smith et al., 2004). As the task in the current study requires both of these functions, we included the N2 component as a dependent variable. We also considered two late ERP components, the N400 and late positive component (LPC), in our analyses. N400 has been found to be sensitive to semantic incongruity in language processing (Kutas & Hillyard, 1980). One study found modulations of N400 amplitude by angry prosody among children (Chronaki et al., 2012). The LPC is sensitive to predictability of words in context (Van Petten & Luka, 2012); and in our study, emotional prosody served as a context of the semantic processing of spoken words. We assumed that the modulation of the LPC depends on how much participants integrate emotional prosody into the semantic context. We expected higher autism trait severity to be associated with smaller congruency effects – indicating prosodic-semantic integration – on accuracy, RT, and early and late ERP components, after controlling for IQ, language ability, and sex.

2. Methods

2.1. Participants

Participants were 46 Cantonese-speaking children (32 boys), aged between 9 and 11.5 years ($M = 10.0$; $SD = 0.76$). They were recruited from schools and organizations which served children with ASD. According to parental reports and clinical reports submitted, 21 children had previously received a diagnosis of ASD or Asperger's syndrome, and nine children were diagnosed with a special need other than ASD, including ADHD ($N = 6$) and dyslexia ($N = 3$). None reported hearing impairment. Autism trait severity was measured using ADOS-2 (Lord et al., 2012); higher ADOS-2 suggests higher autism trait severity or more autistic symptoms. The participants with reported ASD had a significantly higher ADOS-2 scores than those without ASD ($F(1, 44) = 7.96$, $p = .007$). To examine whether there is any confounding effect of non-autistic special needs, a spearman correlation was performed between ADOS-2 score and the presence of non-autistic special needs among participants without ASD. No significant correlation was found ($r = -.131$, $p = .523$).

While in clinical practice an ADOS cutoff serves to identify and confirm ASD diagnoses, we used the ADOS-2 raw total scores as a continuous measure of autism trait severity from absent to severe. This is in line with the RDoC proposed by the U. S. National Institute of Mental Health suggesting that mental disorders should be regarded as the extremes on a continuum. ADOS-2 scores have also been

used as a continuous measure of autism symptom severity in a previous study (Libsack et al., 2022). Intelligence was assessed by a clinical psychologist using the Wechsler Intelligence Scale for Children® – Fourth Edition (Hong Kong short-form [WISC-IV-HK:SF]; Wechsler, 2003) in this study. The mean IQ was 100 ($SD = 17.4$; range = 66 – 132). A speech therapist assessed the participants' language ability using the Hong Kong Cantonese Oral Language Assessment Scale (HKCOLAS) (Child Assessment Service, 2006). Each participant was compensated with HK\$300. Both the children and their legal guardian or an authorized caregiver gave written informed consent to partake in the study. The study procedure was approved by the Research Ethics Committee of Hong Kong Baptist University. All procedures and methods were carried out in accordance with the guidelines and regulations under the Research Ethics Committee.

2.2. Stimuli

The stimuli were disyllabic Cantonese words selected from the “Lexical Lists for Chinese Learning in Hong Kong” (Education Bureau, 2007). Thirty-one children (who did not participate in the ERP experiment) ($Mean\ age = 10.24\ years$; $SD = 1.09$) rated the words' valence and arousal on 5-point Likert scales. Based on these ratings, 60 positive and 60 negative words were selected; none of them were homophones. The valence of the positive and negative words significantly differed ($M = 4.53\ vs.\ 1.46$; $SDs = 0.17\ vs.\ 0.17$; $t(118) = 99.3$, $p < .001$). Negative words were more arousing than positive words ($M = 4.12\ vs.\ 3.07$; $SD = 0.33\ vs.\ 0.51$; $t(103.84) = -12.21$, $p < .001$; degrees of freedom were adjusted for unequal variances between word valence). The lexical frequencies of the stimulus words, representing how often a word is used in daily language, were obtained from a database (Cai & Brysbaert, 2010); the lexical frequencies of positive and negative words did not differ ($t(118) = -.42$, $p = .67$).

The selected words (in equal proportions of negative and positive valence) were spoken by a female or a male professional actor with either angry or happy prosody. Hence 50% of the words were congruent in prosody and word meaning (happy-positive, angry-negative) whereas the other half were incongruent (happy-negative, angry-positive). Rated auditory intensity levels of stimuli did not differ between the four stimulus conditions ($p = .208$). Mean stimulus durations for happy-positive words, happy-negative words, angry-positive words, and angry-negative words were, in order, 857 ($SD = 106$), 867 ($SD = 152$), 650 ($SD = 98$), and 680 ms ($SD = 105$). Mean durations of positive and negative words did not differ ($t(118) = .39$, $p = .70$) but were significantly shorter for angry than happy prosodies ($M = 665\ vs.\ 862\ ms$; $t(238) = 13.02$, $p < .001$). Because speech rate is an essential parameter of emotional prosody, we made no attempt to control for stimulus duration between these conditions.

2.3. Procedure

Participants were seated in a dimly lit, sound-attenuated room in front of a monitor. Words were presented through loudspeakers at the same volume setting for all participants, varying between 60 to 80 dB. Button presses were recorded with a response box (Cedrus® RB-540). Stimulus presentation was administered using Eprime® 2.0 software (PST, Inc.). In each trial, a fixation cross appeared on the screen for 3.1 s. The audio stimuli started 300 ms after the onset of the fixation cross. After the fixation cross disappeared, a feedback screen was shown for 100 ms, with a highly visible red cross or a blank screen indicating an incorrect or correct response, respectively. The feedback was given to ensure better adherence to the task requirement by the children throughout the 240 trials. Stimulus onset asynchronies (SOAs) were equally distributed across values of 3.9, 4.0, and 4.1 s (see Fig. 1).

The children were instructed to fixate on the cross on the computer screen and to classify the valence of the word meaning (positive or negative) as accurately and as quickly as possible by a pressing one of two buttons based on their judgment on whether the meaning was positive or negative, while ignoring the word prosody. Reaction times and response accuracies were recorded as behavioral performance measures. They were also instructed not to blink during stimulus presentation. A total of 240 trials (with 60 trials per stimulus condition) were randomized and organized into six blocks of 40 trials each, with breaks in between. To familiarize the participants with the task, they were given eight practice trials.

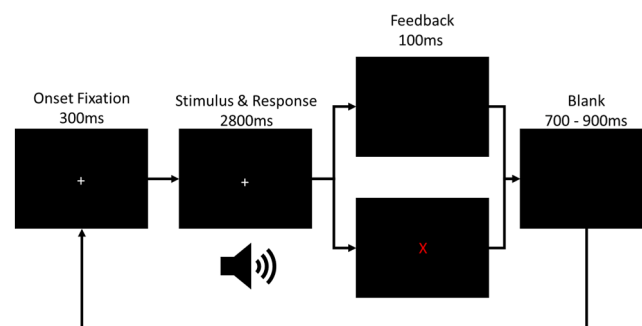


Fig. 1. Illustration of procedure of prosody task.

2.4. Electroencephalogram (EEG) recordings and pre-processing

The EEG was recorded with 41 cap-mounted Ag/AgCl electrodes (Waveguard™ original; initial reference: CPz; impedance < 20 kΩ) connected to an eego™mylab (ANT Neuro) amplifier and sampled at 1000 Hz. Offline processing was performed with the EEGLAB toolbox and in-house MATLAB scripts. EEG data had a bandpass at 0.01 – 40 Hz and were re-referenced to a common average. The continuous EEG was epoched from 200 ms pre- to 2000 ms post-stimulus onset and baseline-corrected against the 200 ms pre-stimulus interval. Artifacts due to ocular and body movements were removed by an ICA algorithm in the EEGLAB toolbox. Epochs with residual artifacts were removed by visual inspection. The average number of rejected trials were 2.8 (SD = 4.7; Range: 0 to 19) due to artifacts and 31.6 (SD = 18.5; Range: 9 to 125) due to incorrect responses, out of the 240 trials in total per person.

Finally, we applied Residue Iteration Decomposition (RIDE) (Ouyang et al., 2011, 2015), which reconstructs the ERP after correcting for trial-by-trial latency jitter. Since we used disyllabic words as stimuli, the recognition point of semantic meaning varied from word to word (Marslen-Wilson, 1987), as did ERP latencies in response to these stimuli, inducing jitter and smearing of ERP waveforms. RIDE decomposes the ERP into stimulus-locked, response-locked, and a non-time-locked component clusters. By taking latency variability into account, it realigns the component clusters and reconstructs the ERPs, synchronized at their respective latencies.

2.5. EEG data analyses

To identify clusters of electrodes that are highly correlated, spatial principal components analysis (PCA) was performed on the RIDE-reconstructed ERP data for all electrodes, time points (0–2000 ms), and conditions, using in-house MATLAB scripts. Spatial PCA captures the intercorrelated electrodes and linearly combines the electrodes with weightings to form a virtual component. PCA components were Varimax rotated (Spencer et al., 2001) and variables with factor loading > .7 were selected. Factor scores were formed by taking the dot product of site values multiplied by respective factor score coefficients (Spencer et al., 2001). For each participant, ERP waveforms were synthetically reconstructed for each spatial component based on the factor scores. The spatial PCA has an advantage over traditional ERP analysis methods as it captures the relevant variance, preserving the information from the electrodes, and at the same time reducing the redundancy among them (Spencer et al., 2001).

We extracted the N2, P2, N400, late negativity, and LPCs using conventional time windows and with reference to the post-PCA grand average ERP waveforms. ERP amplitudes were calculated using average values within a ± 50 ms window around local peaks of the components. The time window for searching of the N2 peak was 100 – 300 ms (Folstein & Van Petten, 2008). P2 was located in the 100 – 300 ms time window (Bourisly & Shuaib, 2018), and N400 within the 350 – 500 ms time window. Visual inspection of the grand average shows a late fronto-central negativity and a late parieto-occipital positivity component. The search windows for these components were 500–900 ms and 800–1200 ms, respectively, covering the peaks in the grand average.

2.6. Statistical analyses

For behavioral data, separate ANCOVAs were performed on accuracies and RTs as dependent variables, with prosody and congruence as repeated measures factors, sex as between-subject factor, and IQ, ADOS, and language ability as covariates, with the SPSS Statistics 25 software. Please note that semantic valence of stimulus words was not included in the analysis as it is not a concern in our research objectives. RTs were calculated based on correct trials only.

For EEG data, PCA extracted two spatial components (see Section 3.2 ERP Results for details). The distributions of ERP amplitudes for the two spatial components in different time windows were exported to SPSS Statistics 25 software for further analyses. Repeated

Table 1
Sample statistics and semantic valence task performance.

			Male (n = 32)				Female (n = 14)			
			M	SD	Min	Max	M	SD	Min	Max
Sample statistics										
Age			10.0	.83	9.0	11.5	10.1	.62	9.0	11.0
HKCOLAS			11.6	3.78	0.0	18.0	11.9	2.64	8.0	17.0
IQ			98.8	17.42	66.0	127.0	104.5	17.21	73.0	132.0
ADOS raw total score			4.2	3.61	0.0	12.0	2.9	3.20	0.0	12.0
Semantic valence task performance										
Happy Prosody	Accuracy	Overall	.859	.09	.46	.97	.911	.041	.81	.97
		Congruent	.877	.10	.50	1.00	.921	.040	.85	.98
		Incongruent	.840	.09	.42	.98	.899	.060	.73	.97
	RT (ms)	Overall	1375	114	1087	1578	1353	92.9	1230	1524
		Congruent	1334	110	1029	1524	1290	100.3	1169	1464
		Incongruent	1416	128	1145	1635	1417	96.8	1291	1602
Angry Prosody	Accuracy	Overall	.842	.84	.48	.94	.894	.040	.81	.96
		Congruent	.886	.10	.48	1.00	.952	.046	.82	1.00
		Incongruent	.798	.10	.48	.92	.838	.042	.77	.92
	RT (ms)	Overall	1257	122	923	1436	1252	105	1080	1447
		Congruent	1268	138	875	1471	1255	110	1062	1449
		Incongruent	1245	112	971	1409	1249	110	1099	1502

measures ANCOVA was performed on the amplitudes for each ERP component as dependent variable. Prosody and congruence were repeated measures factors, sex was a between-subject factor, and IQ, ADOS, and language ability were covariates. Post hoc regression analyses were performed to study patterns of associations between variables as appropriate.

3. Results

3.1. Behavioral results

Descriptive statistics for age, ADOS raw total scores, language ability, and IQ are provided in Table 1. Independent sample *t*-tests showed no significant sex differences in ADOS scores, language ability, IQ, or age ($ps > .05$). Please note the negative correlations between ADOS scores and language ability ($r = -.551, p < .001$), IQ ($r = -.452, p = .002$), and performance accuracy ($r = -.426, p = .003$). Further correlations are reported in Table 2.

Separate ANCOVAs were performed on accuracy and RTs as dependent variables, with prosody and congruence as repeated measures factors, sex as between-subject factor, and IQ, ADOS, and language as covariates. Post hoc regression analysis was used to explore effects of ADOS on RT. Please note that word semantic valence was not included in the analysis as it is not a concern in our research objectives. RTs were calculated based on correct trials only.

Higher accuracy was associated with femaleness ($F(1, 41) = 4.36, p = .043$, partial $\eta^2 = .096$), higher IQ ($F(1, 41) = 5.15, p = .029$, partial $\eta^2 = .11$; Fig. 2a) and higher language ability ($F(1, 41) = 6.66, p = .014$, partial $\eta^2 = .14$; Fig. 2b).

There were significant associations of both IQ ($F(1, 41) = 7.30, p = .010$, partial $\eta^2 = .15$) (Fig. 2c) and ADOS ($F(1, 41) = 6.60, p = .014$, partial $\eta^2 = .14$) (Fig. 2d) with RTs, and a significant interaction between congruence and sex ($F(1, 41) = 4.25, p = .046$, partial $\eta^2 = .094$). A post hoc regression analysis with IQ, ADOS, language, and sex as predictors showed that higher ADOS ($\beta_{\text{standardized}} = -.42, SE = 5.05, t = -2.57, p = .014, 95\% \text{ CI of } \beta_{\text{standardized}}: -.74 \text{ to } -.089, f^2 = .13$) and higher IQ ($\beta_{\text{standardized}} = -.44, SE = 1.02, t = -2.70, p = .010, 95\% \text{ CI of } \beta_{\text{standardized}}: -.077 \text{ to } -.11, f^2 = .14$) were associated with faster RTs (Fig. 2c and d).

3.2. ERP results

Fig. 3 illustrates the grand average ERPs at Fz, Cz, and Pz electrodes before and after applying RIDE. Note that the ERP deflections prior to word onset at time 0 are likely visually evoked potentials triggered by the fixation cross, which appeared at -300 ms.

The PCA extracted two spatial components (accounting for 96.0% of variance): fronto-central (88.2%) and parieto-occipital (7.8%) channels. The fronto-central component consisted of 19 electrodes (F3, F4, F7, F8, FC1, FC2, FC5, FC6, Fz, FP1, FP2, FPz, C3, C4, CP1, CP5, Cz, T7, and T8), and the parieto-occipital consisted of 9 electrodes (CP2, CP6, P3, P4, P7, P8, POz, Pz, and O2). The topographic distributions of ERP amplitudes for the two spatial components in different time windows are shown in Fig. 4. Fig. 4a and b show grand average ERPs for each spatial component in congruent and incongruent conditions.

The same type of repeated measures ANCOVA was performed on the ERP amplitudes for each component as dependent variable. Prosody and congruence were repeated measures factors, sex was a between-subject factor, and IQ, ADOS, and language were covariates. Post hoc regression analyses were performed to study associations between variables as appropriate.

3.2.1. Fronto-central spatial component

3.2.1.1. N2 (100–300 ms). Larger N2 amplitudes were elicited by angry than by happy prosodic stimuli ($M = -2.65$ vs. -2.58) ($F(1, 41) = 4.62, p = .038$, partial $\eta^2 = .10$; Fig. 4). The interaction between prosody and language ability was significant ($F(1, 41) = 6.24, p = .017$, partial $\eta^2 = .13$).

3.2.1.2. Late negativity (500–900 ms). Angry prosodic stimuli ($M = -7.41$) elicited more negative amplitudes than happy prosodic stimuli ($M = -6.68$) ($F(1, 41) = 4.38, p = .043$, partial $\eta^2 = .10$; Fig. 4).

Table 2

Bivariate correlations between all behavioral variables ($N = 46$).

		ADOS	HKCOLAS	IQ	Accuracy	Reaction Time
1	Age	-.073	.041	-.142	.085	-.018
2	ADOS		-.551 ***	-.452 **	-.426 **	-.052
3	HKCOLAS			.556 ***	.589 ***	-.306 *
4	IQ				.579 ***	-.418 **
5	Accuracy					-.011

*** $p < .001$, ** $p < .01$, * $p < .05$

Note. IQ = WISC-IV score; ADOS = ADOS-2 raw total score; HKCOLAS = Language comprehension score; Accuracy = accuracy of the semantic valence judgment of words; Reaction Time = reaction time of the semantic valence judgment of words

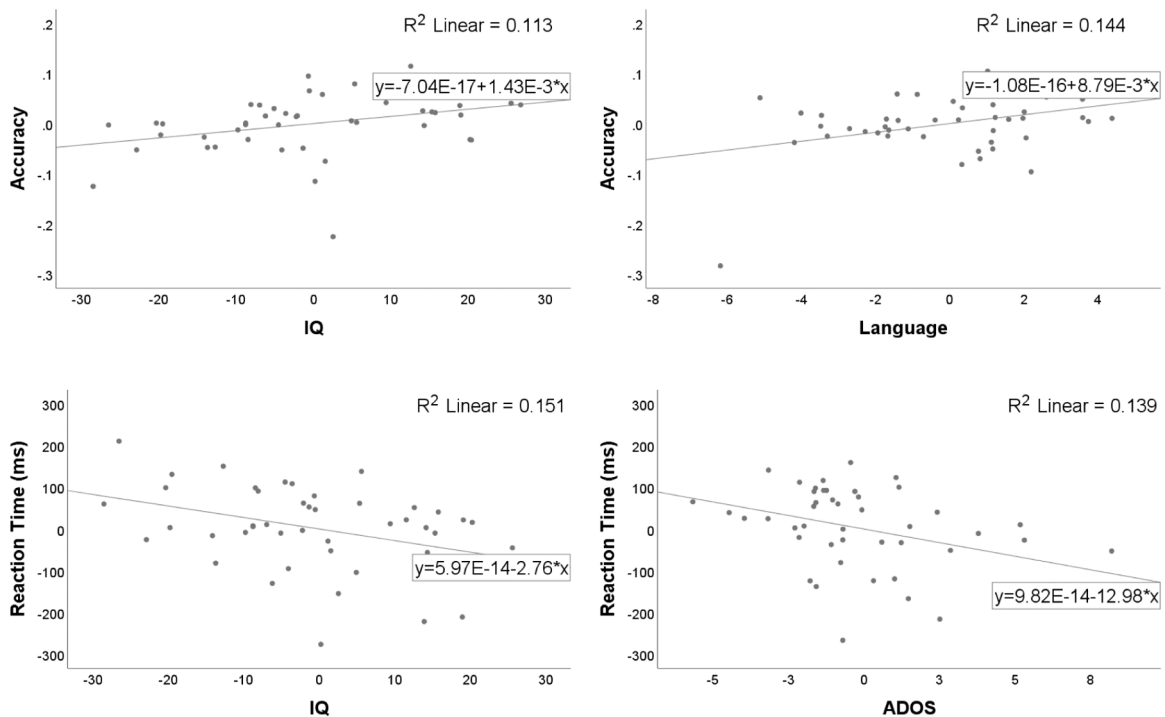


Fig. 2. 2a (top left), 2b (top right), 2c (bottom left) and 2d (bottom right). Performance in classifying positive and negative word meaning as a function of IQ (left), language ability (top right), and ADOS score (bottom right). Please note that the data are mean-centered.

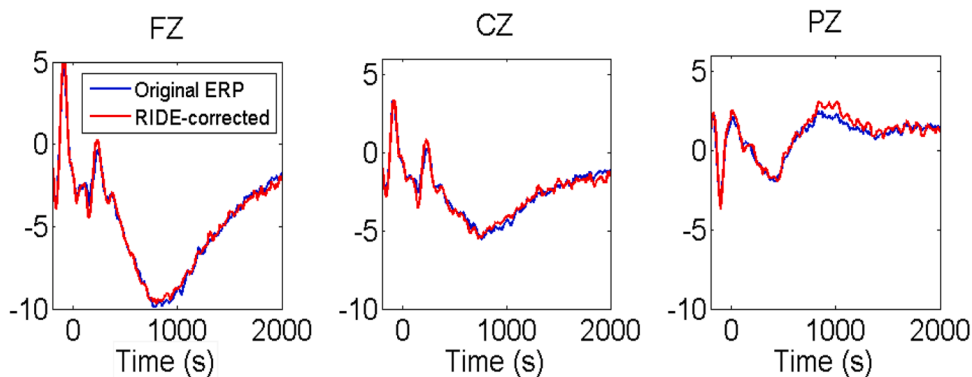


Fig. 3. Grand average ERP waveforms at Fz, Cz, and Pz electrodes before and after RIDE reconstruction. Time 0 is the onset of the auditory word stimuli.

3.2.2. Parieto-occipital spatial component

3.2.2.1. P2 (100–300 ms). Angry prosodic stimuli ($M = .28$) elicited more positive P2 amplitudes than happy prosodic stimuli ($M = .10$) ($F(1, 41) = 5.37, p = .026$, partial $\eta^2 = .12$; Fig. 4). There were significant interactions between prosody and ADOS ($F(1, 41) = 6.67, p = .014$, partial $\eta^2 = .14$), prosody and congruence ($F(1, 41) = 6.57, p = .014, \eta^2 = .14$), and between congruence and sex ($F(1, 41) = 7.70, p = .008$, partial $\eta^2 = .16$). There was also a three-way interaction of prosody $\times$ congruence $\times$ ADOS ($F(1, 41) = 4.58, p = .038$, partial $\eta^2 = .10$). To examine the direction of the association between continuous variables and the dependent variables in the interactions mentioned, two post hoc regressions were conducted with ADOS, IQ, sex, and language as control variables, and the P2 congruency effects (differences between incongruent and congruent condition) of happy and angry prosodic stimuli as dependent variables. The overall model for the angry prosodic condition was significant ($F(4, 41) = 4.19, p = .006, R^2 = .29$). Higher ADOS ($\beta_{\text{standardized}} = -.33, t = -2.03, p = .049$, 95% CI of $B_{\text{standardized}}$: $-.66$ to $-0.00, f^2 = .076$) and higher IQ ($\beta_{\text{standardized}} = -.42, t = -2.58, p = .013$, 95% CI of $B_{\text{standardized}}$: $-.75$ to $-0.09, f^2 = .13$) were associated with a smaller P2 congruency effect (Fig. 5). Boys showed a smaller P2 congruency effect than girls ($\beta_{\text{standardized}} = -.36, t = -2.71, p = .010$, 95% CI of $B_{\text{standardized}}$: $-.64$ to $-0.09, f^2$

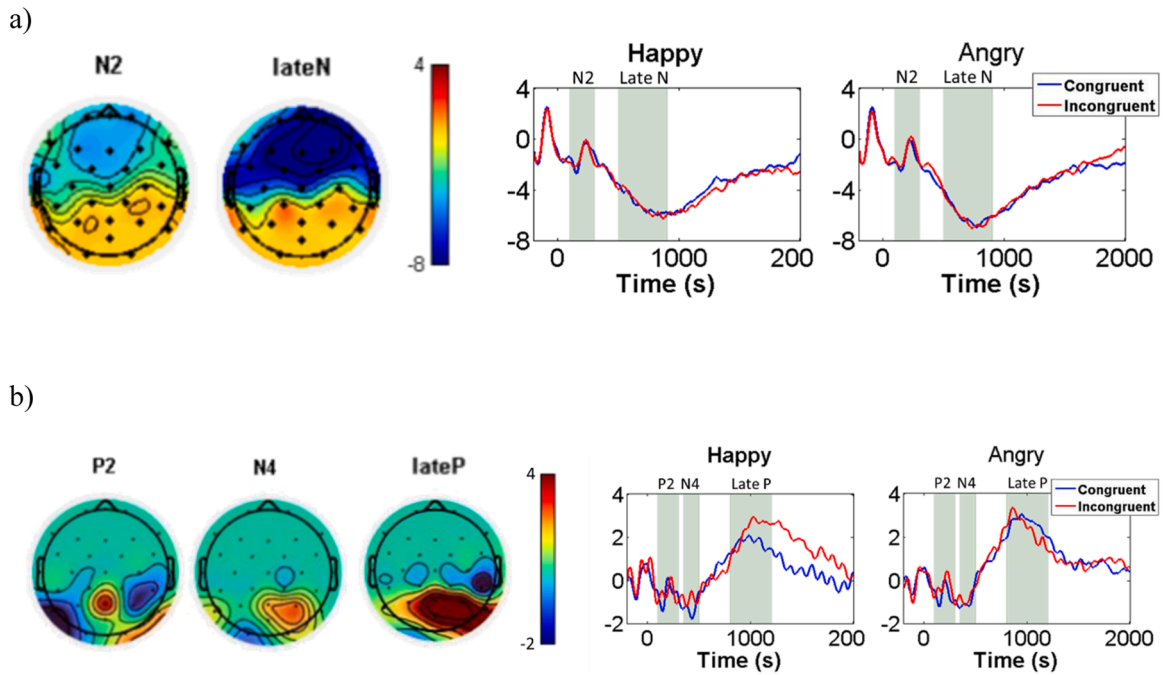


Fig. 4. Topographies of ERP components (left) and grand average ERPs (right) for each condition. (a) Fronto-central spatial component. The shaded regions are the N2 (100 – 300 ms) and the late negativity (500 – 900 ms) components respectively. (b) Parieto-occipital spatial component. The shaded regions are the P2 (100 – 300 ms), the N4 (350 – 500 ms), and the late positivity (800 – 1200 ms) respectively. A 10-Hz low-pass filter has been applied for illustration only.

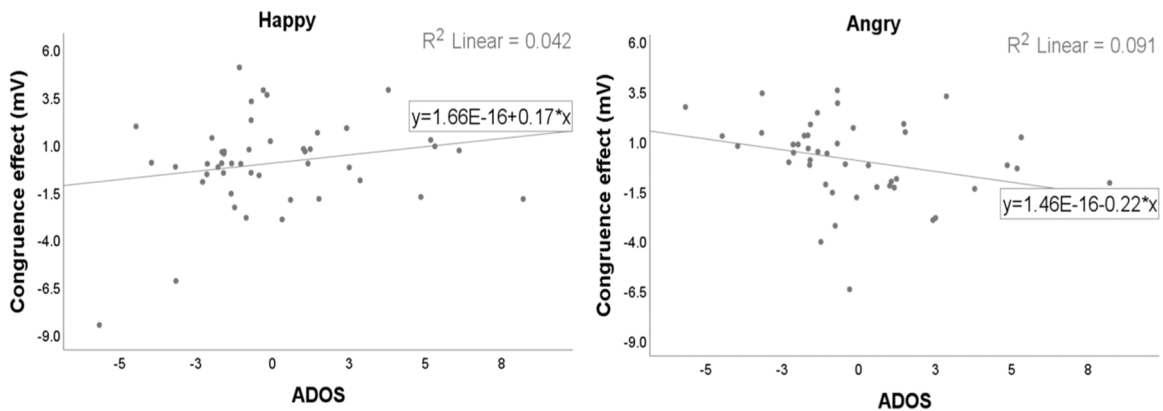


Fig. 5. The partial regression plots of the relationship between the congruence effect in the parieto-occipital P2 and ADOS scores for happy (left) and angry (right) prosodic condition. Please note that the data are mean-centered.

=.15). The overall regression model for the happy prosodic condition was not significant ($F(4, 41) = .92, p = .46, R^2 = .08$).

3.2.2.2. N400 (350–500 ms). Happy prosodies elicited larger N400 amplitudes than angry prosodies ($M = -1.50$ vs. -1.35) ($F(1, 41) = 16.81, p < .001$, partial $\eta^2 = .29$; Fig. 4). There were significant interactions between prosody and ADOS ($F(1, 41) = 9.26, p = .004$, partial $\eta^2 = .18$), and between prosody and IQ ($F(1, 41) = 9.18, p = .004$, partial $\eta^2 = .18$). To study the patterns of associations, two post hoc regressions were conducted with ADOS, IQ, sex, and language ability as independent variables, and N400 amplitudes of happy and angry prosodies, respectively, as dependent variables. The overall model for happy prosodic stimuli was significant ($F(4, 41) = 3.83, p = .010, R^2 = .27$). Higher ADOS was associated with smaller N400 amplitudes for happy prosodic stimuli ($\beta_{\text{standardized}} = .39, t = 2.39, p = .022$, 95% CI of $\beta_{\text{standardized}}: .06$ to $.73, f^2 = .11$). Boys showed smaller N400 amplitudes than girls for happy prosodic stimuli ($\beta_{\text{standardized}} = .29, t = 2.15, p = .037$, 95% CI of $\beta_{\text{standardized}}: .02$ to $.57, f^2 = .089$). No significant effect was found for the angry prosodic condition was not significant ($F(4, 41) = 2.29, p = .076, R^2 = .18$).

3.2.2.3. LPC (800–1200 ms). Overall, boys showed larger LPCs than girls ($F(1, 41) = 4.60, p = .038$, partial $\eta^2 = .10$) and there was significant interaction between congruence and sex ($F(1, 41) = 4.73, p = .036$, partial $\eta^2 = .10$). A post hoc regression showed that the congruency effect (difference between incongruent and congruent condition) was larger for girls than boys ($t = -2.17, p = .036, R^2 = .18$).

4. Discussion & implications

4.1. General discussion

The present study demonstrates for the first time that children with higher autism trait severity respond faster (without trading accuracy) when judging the semantic valence of emotionally spoken words, after controlling for IQ, language ability, and sex. A study by Harris et al. required children to judge the valence (positive vs. negative) of visually presented words (Harris et al., 2006). Although they found similar performance between ASD individuals and controls, the ASD group showed smaller fMRI activation at the Broca's area and larger at the Wernicke's area than the control group, suggesting recruitment of different neural networks to accomplish the task. One difference between the present study and that of Harris and colleagues' is the use of spoken instead of written words as stimuli. It is possible that children with higher autism trait severity may respond faster because they tend to ignore task-irrelevant prosodies, as discussed next.

In the ERP data, higher autism trait severity and higher IQ went along with smaller congruency effects at P2 for angry prosodic stimuli, which implies less automatic early prosodic-semantic integration of angry stimuli. Less automatic prosodic processing in ASD was also found in Lindström et al., (2016, 2018), in which children with ASD showed smaller MMN and P3a responses to scornful deviants. Alternatively, this can also be interpreted as a greater ability to ignore or inhibit task-irrelevant angry prosodies, which is supported by a past fMRI study which found heightened activities in cognitive-control brain areas when processing angrily spoken sentences among adolescents with ASD (Eigsti et al., 2012). Since smaller congruency effects were only found for angry but not happy prosodic stimuli in our study, the findings support the empathy imbalance theory, which suggests that the hyper-aroused emotional systems in ASD are linked to avoidant behaviors to restrict excessive arousal triggered by intense and negative emotions (Smith, 2009). With fMRI, Zalla and Sperduti (2013) showed that aversive stimuli overstimulate the amygdala of individuals with ASD, due to defective top-down prefrontal regulation. This may explain why angry (but not happy) prosody was disattended in our study. Consistent with this, García-Blanco et al. (2017) also found avoidance of angry but not happy faces in individuals with ASD.

Another novel finding is that the children with higher IQ demonstrated less early automatic processing of angry prosody, or better inhibition of angry task-irrelevant prosody. The ability to inhibit irrelevant information is an important executive function typically associated with higher IQ (Polderman et al., 2009). Please note that we did not observe similar effects in the happy prosodic words. Previous studies have found different patterns of top-down emotion regulation when individuals process positive and negative emotional stimuli (Deng et al., 2013). It is therefore not surprising that the effect of IQ on the P2 congruency effect was found only in the angry but not the happy prosodic condition.

Please note that while the correlation between ADOS and IQ was negative, these two variables were both positively associated with the P2 congruency effect in the regression model. In regression, the association between one predictor and the dependent variable is calculated when the other predictors in the model are held constant. In other words, our findings imply that in a group of children with the same IQ, language ability, and sex, those with higher ADOS had smaller P2 congruency effect.

Significant association of autism trait severity were also found in a later ERP component, N400, in response to happy but not angry prosodic stimuli: Children with high autism trait severity showed smaller N400 amplitudes in response to happy prosodic stimuli as compared to children with low autism trait severity. The N400 component has been linked to semantic memory retrieval, expectancy of words and the integration of word meaning into context (Kutas & Federmeier, 2011). While further investigation will be needed to reveal the mechanisms behind these associations, the presence of these relationships in happy but not angry stimuli may indicate that children with high autism trait severity have low expectancy towards happy stimuli while holding similar expectancies for angry stimuli, compared to children with lower autism trait severity.

Angry prosodies elicited larger amplitudes in the N2 and late negativity than happy prosodic stimuli in the fronto-central spatial component. The fronto-central N2 has been consistently linked with executive cognitive control, specifically response inhibition (Nieuwenhuis et al., 2003) and conflict monitoring (van Veen & van Veen & Carter, 2002). Moreover, the N2 has been related to emotion regulation in children (Lewis et al., 2006) and studies show a larger N2 amplitude in response to angry faces than to happy faces (Lewis et al., 2007). Lewis and colleagues (2007) proposed that the increased N2 reflects increased attention because of the saliency of angry stimuli, the regulation of anxiety triggered by angry faces, or the effort to inhibit an action because an angry face often indicates a stop signal from parents or adults (Blair et al., 1999). The children in our study might have recruited more mental resources to regulate their anxiety when hearing the angry voices of the professionally trained speakers.

The late negativity has been related to post-stimulus evaluation of response selection and conflict monitoring (Berti, 2008). Such late negative components have been associated with the congruency between vocal cues and speech context in adults (Schirmer & Kotz, 2003; Schirmer et al., 2006). It appears that more mental resources are allocated to response conflict monitoring in the angry than the happy prosodic condition. Behaviorally, accuracy in the angry prosodic condition was lower, which may imply greater task difficulty in the semantic processing of angrily spoken words and more mental resources being drawn on for response conflict monitoring.

The prosody effect was somewhat different at the two parieto-occipital P2 and N400 components: angry prosodic stimuli elicited larger P2, but smaller N400 amplitudes than happy prosodic stimuli. The P2 has been linked to early attentional processes to emotional

stimuli, where negative stimuli elicited a larger P2, as compared to positive stimuli (Hashimoto et al., 2012). While angry stimuli appear to recruit more attention (indicated by larger P2), they may be more expected or easier to retrieve from semantic memory (indicated by smaller N400) than happy stimuli.

Finally, there were no main effects of congruence in either behavioral or ERP results. This is at variance with findings in adults (Schirmer et al., 2006; Lui et al., 2020) and also in children, which show such integration effects. Friend (2000) showed congruence effect between emotional prosody and lexical word content in girls aged 4, 7, and 10 years. One possible reason for the discrepancy with the present results could be gender-related – the participants were predominantly boys in the present study and all girls (Friend, 2000). The girls in our study showed larger congruency effects in reaction times and accuracy than the boys, when controlling for language ability, IQ, and autism trait severity. Hence, the girls more strongly integrated emotional prosody into word meaning, which they also processed more quickly than the boys. This pattern is consistent with studies on adults (Schirmer et al., 2002, 2006). The lack of a main effect of congruence in our study can also be attributed to the characteristics of our children participants with high autism trait severity. This is in line with our finding that children with high autism trait severity show smaller congruency effects in the P2 component.

4.2. Limitations

Our study recruited mainly boys (70%; 32 out of 46) as participants because we aimed to include children with high autism trait severity, and males are on average higher in autistic traits (Baron-Cohen & Wheelwright, 2004). Although sex was included in the analysis, no interpretation on sex difference was attempted because of the imbalanced sample sizes. It is noted that sex differences are not the focus of our study and the factor of sex was included to partial out its potential confounding effect.

The participants' autism trait severity was assessed using the ADOS-2 Module 4, which had been validated in a sample of 393 individuals aged between 10 and 62 years (Hus & Lord, 2014). According to the administration manual of ADOS-2 (Lord et al., 2012), "the major differences between modules 3 and 4 lie in the greater use of toys in Module 3 and a greater emphasis on interview questions in Module 4." It was also suggested in the manual that the decision to use Module 3 versus Module 4 should be dependent on the client's maturity level and interest in toys. Our participants were verbally fluent adolescents aged between 9 to 12 years, and by the clinical psychologist's judgment, Module 4 is more appropriate for our participants based on their expressive language ability and maturity level. Moreover, given that our objective was to examine a continuous level of severity instead of dichotomous group differences, we believe that measuring autism trait severity using ADOS-2 module 4 served our purpose well.

For stimulus construction, the emotionally spoken words were recorded by a professional actor and an actress. Although this method of stimulus construction is widely used in emotion research, there could be a concern about ecological validity as those stimuli are not recorded from natural settings in which people spontaneously produce emotional tones of voice while experiencing certain emotions. There are, however, feasibility issues in obtaining recordings of real-life emotional prosodic stimuli. Moreover, stimulus characteristics such as duration and pitch level varied between prosodic conditions. It is uncertain how the stimulus differences contributed to the differential patterns of ERP findings across conditions.

Finally, the meanings of negatively valenced words used in our study could link to different kinds of negative emotions, such as sadness, disgust, and fear. Therefore, there may not be a perfect congruency between semantic meaning and emotional prosody for the angrily spoken negative words (e.g., a word with sad meaning spoken in an angry tone). It is, however, noted that the level of congruency between sadness and anger is still higher than that between happiness and anger. It is evidenced in previous studies (Schirmer & Kotz, 2003) that the congruency effect was much higher for happily spoken negative words compared to angrily spoken negative words, even though the negative word meanings are not necessarily related to anger but other negative emotions.

4.3. Conclusion and implications

In conclusion, we have shown here that children with high autism trait severity may have less automatic emotional prosodic-semantic integration of angrily spoken words, and the behavioral pattern could be linked to the early stages of attentional processes as shown by the ERP findings. One should note that less automatic processing of task-irrelevant emotional signals can sometimes be an asset rather than a deficit, as ignoring irrelevant emotional signals may help individuals to allocate mental resources to perform more efficiently in some situations. For example, rescuing team and firemen may need to inhibit emotions to stay focused on the problem solving when facing disasters or emergency situations.

There is another practical implication for parents and educators regarding the finding of the ignorance of angry prosodies among children with autism trait severity. Caregivers and educators may benefit from the awareness the use of angry prosodies may not have the same impact on children with high autism trait severity as compared to children with low autism trait severity.

Finally, our study is novel in that it examined the continuum of autistic traits among a population of children including both clinical and non-clinical cases, instead of carrying out group comparisons. This approach is consistent with the Research Domain Criteria (RDoC) proposed by the U.S. National Institute of Mental Health (NIMH), which suggests that mental disorders should be regarded as the extreme ends on a continuum spanning the entire population of healthy to malfunctioning individuals. Further studies may continue to explore associations with autistic traits and behavioural and physiological phenotypes among both clinical and non-clinical populations with the dimensional perspective emphasized by RDoC.

CRedit authorship contribution statement

Gilbert Ka-Bo Lau: Formal analysis, Resources, Data curation, Writing – original draft preparation. **Ming Lui:** Conceptualization, Project administration, Funding acquisition, Methodology, Investigation, Writing – original draft preparation. **Andrea Hildebrandt & Werner Sommer:** Writing – review & editing. **Changsong Zhou:** Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors report no conflicts of interest.

Data availability

Data will be made available on request.

Acknowledgements

This work was supported by the Hong Kong Baptist University Research Committee Interdisciplinary Research Matching Scheme [RC- IRMS/16–17/04]; HKBU Interdisciplinary Research Clusters 2018/19 [RC-IRCMs/18–19/SCI/01]; HKBU Faculty Niche Research Area Initial Grant [RC-FNRA-IG/20–21/SCI/04]; and the General Research Fund of Research Grants Council, Hong Kong [Ref No. 12604418]. The authors declare no conflicts of interest.

Data Availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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